

Feynmann Rules in xspace & pspace:

Proof of Rules:

$$G(x_1, \dots, x_n) = \frac{\delta^n Z[\mathcal{J}]}{\delta \mathcal{J}_{x_1} \dots \delta \mathcal{J}_{x_n}} \Big|_{\mathcal{J}=0} = e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}} \left\{ \phi_{x_1} \dots \phi_{x_n} e^{-\int d^d x V(\phi) + \mathcal{J} \phi} \right\}_{\mathcal{J}=0, \phi=0}$$

$$G(x_1, \dots, x_n) = e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}} \left\{ \phi_{x_1} \dots \phi_{x_n} e^{-\int d^d x V(\phi)} \right\}_{\phi=0}$$

$$\text{Free theory: } \langle 0 | \dots | 0 \rangle = e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}} \left\{ \phi_{x_1} \dots \phi_{x_n} e^{-\int d^d x V(\phi)} \right\}$$

Want: Interacting theory:
 $\langle \Omega | \phi_1 \dots \phi_n | \Omega \rangle$

* I can drop $\mathcal{J} \phi$ terms since the terms we obtain from it have $\mathcal{O}(\mathcal{J})$ and will vanish as

$$\mathcal{J} \rightarrow 0$$
$$G(x_1, \dots, x_n) = e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}} \left\{ \phi_{x_1} \dots \phi_{x_n} e^{-\int d^d x V(\phi)} \right\}_{\phi=0}$$

* To order N : $\mathcal{O}(\lambda^N)$:

$$G^{(N)}(x_1, \dots, x_n) = e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}} \left\{ \phi_{x_1} \dots \phi_{x_n} \frac{(-\lambda)^N}{N!} \int d^d y_1 \dots d^d y_N \phi_{y_1}^{p_1} \dots \phi_{y_N}^{p_N} \right\}$$

Expand $e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}}$ to order $n + pN$

Recall:

$$V(\phi) = \lambda \phi^p$$

* We need there to be as many derivatives as there are ϕ^* s on the RHS.

$$\boxed{2q = n + pN}$$

q : order of $e^{\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi}}$

n : $\phi_{x_1} \dots \phi_{x_n}$ in front

N : order of the pieces in $G^{(N)}$

$$\frac{1}{q!} \left(\frac{1}{2} \frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi} \right)^{q!} \left\{ \phi(x_1) \dots \phi(x_n) \int d^d y_1 \dots d^d y_n \phi^p(y_1) \dots \phi^p(y_n) \right\}_{\phi=0}$$

$$\frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi} = \int d^d z_i d^d w_i \left(\frac{\delta}{\delta \phi(z_i)} \Delta(z_i - w_i) \frac{\delta}{\delta \phi(w_i)} \right)$$

$$* \left(\frac{\delta}{\delta \phi} \Delta \frac{\delta}{\delta \phi} \right)_{z_i, w_i} (\phi(x_i) \phi(x_j))$$

$$= \int d^d z_i d^d w_i \frac{\delta}{\delta \phi(z_i)} \Delta(z_i - w_i) \left(\delta(w_i - x_j) \phi(x_i) + \delta(w_i - x_i) \phi(x_j) \right)$$

$$= \int d^d z_i d^d w_i \Delta(z_i - w_i) \left\{ \delta(w_i - x_i) \delta(x_j - z_i) + \delta(w_i - x_j) \delta(z_i - x_i) \right\}$$

$$= \int d^d z_i \left\{ \Delta(z_i - x_i) \delta(x_j - z_i) + \Delta(z_i - x_j) \delta(z_i - x_i) \right\}$$

$$= \Delta(x_j - x_i) + \Delta(x_i - x_j)$$

$$= 2 \Delta(x_i - x_j) = 2 \Delta_{ij}$$

• Acting on all of the RHS 2^q terms and so the 2^q terms cancel. There are $q!$ permutations which also cancel.

Statistical weight fac for:

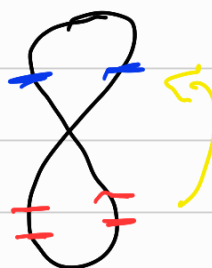
• Define $\lambda \equiv \lambda_p / p!$ to cancel most factors

$$G^{(n)} \sim \frac{1}{S} ()$$

$$S = \frac{N! (p!)^N}{\# \text{ equivalent diagrams}}$$

$$S_{\text{tree}} = 2$$

(i)  has $S = 8$



$$2 \times 2 \times 2 = 8$$

(ii)  $S = 2$



(i)  $S = 3!$

Feynman Rules in p-space:

$$\tilde{G}(p_1, \dots, p_n) = \int \prod_i d^d x_i e^{i \sum_j x_j p_j} G(x_1, \dots, x_n)$$

$$= (2\pi)^d \delta(p_1 + \dots + p_n) \underbrace{G(p_1, \dots, p_n)}$$

momenta already satisfy
momentum conservation.

$$\Delta(p) = \frac{1}{p^2 + m^2}$$

• Momentum going out of x : $e^{i p x}$
Momentum in to x : $e^{-i p x}$

 $e^{i p x}$

 $e^{-i p x}$

ex: $\tilde{G}(p_1, p_2) = \int d^d x_1 d^d x_2 e^{i p_1 x_1 + i p_2 x_2} G(x_1, x_2)$

$$= \int d^d x_1 d^d x_2 e^{i p_1 x_1 + i p_2 x_2} \int \frac{d^d p}{(2\pi)^d} e^{i p (x_1 - x_2)} \frac{1}{p^2 + m^2}$$

$$= \int \frac{d^d p}{(2\pi)^d} \int d^d x_1 d^d x_2 e^{i (p + p_1) x_1} e^{i x_2 (p_2 - p)} \frac{1}{p^2 + m^2}$$

$$= \int d^d x_1 \left[\int d^d p \delta(p_2 - p) e^{i (p + p_1) x_1} \frac{1}{p^2 + m^2} \right]$$

$$= \int d^d x_1 e^{i (p_2 + p_1) x_1} \left[\frac{1}{p_2^2 + m^2} \right]$$

$$= (2\pi)^d \delta(p_1 + p_2) \underbrace{\frac{1}{p_2^2 + m^2}}_{G(p_2)}$$

$\tilde{G}(p_1, p_2) = (2\pi)^d \delta(p_1 + p_2) G(p_2)$